Energy and Power for FTC Robots

Every FTC robot runs on a 12 V NiMH battery pack rated at 3000 mAh. This section you will use what you learned about current, voltage, charge, energy and power to answer two practical questions. How much of the battery does a mechanism use? And why does the battery's voltage drop as it runs down?

Energy From a Motor at Constant Current

Power is the rate at which energy moves through a circuit. For any electrical load,

$$\mathbf{P = I \times V}$$

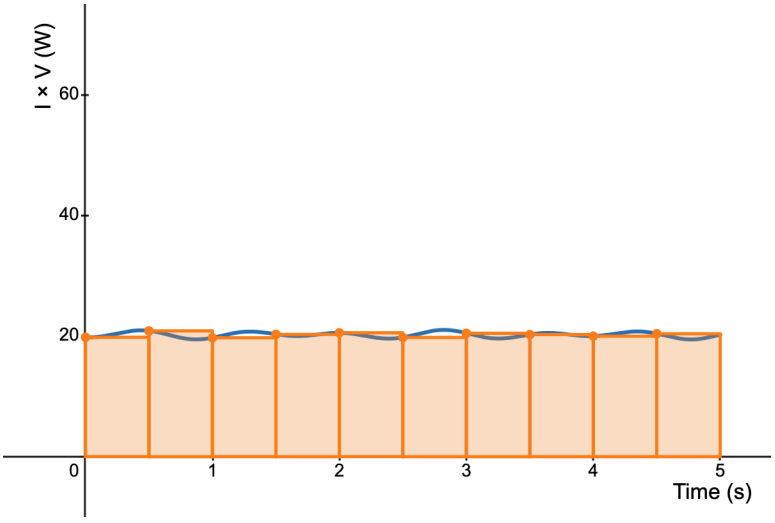
where P is power in watts (W), I is current in amps (A), and V is voltage in volts (V).

If the power stays the same for a length of time Δt, the energy used is the power multiplied by that time:

$$\mathbf{E = I \times V \times \Delta t}$$

The units used for time tell you what units you get for the energy. If Δt is in seconds, E is in joules (J), because 1 W = 1 J/s. If Δt is in hours, E is in watt-hours (Wh). One watt-hour is 3,600 J.

You can use a graph of **V × I** vs. **time** to visualize this calculation. Each rectangle on the graph below has an area of **(V × I) × Δt** where Δt in this case is 0.5 s. The total energy is the sum of the areas for all 10 rectangles.



In the case of a constant current this gives the same result as the equation E = V × I × Δt. In this case E = 12 V × 1.67 A × 5 s = 100 J. Running the device longer is the same as adding more rectangles to the graph which means more energy is used.

Example. A drivetrain motor draws a steady 2.0 A from a 12.0 V battery for a full 150 s match.

- Power: $P = I \times V$

  $P = 2.0\ A \times 12.0\ V$

  $P = 24\ W$

- Energy: $E = P \times \Delta t$

  $E = 24\ W \times 150\ s$

  $E = 3{,}600\ J$

  or: $E = 3{,}600\ J \times \frac{1\ h}{3600\ s} = 1\ Wh$

- Charge: $Q = I \times \Delta t$

  $Q = 2.0\ A \times 150\ s$

  $Q = 300\ J$

  Or: $Q = 300\ J \times \frac{1\ h}{3600\ s} = 0.083\ Ah = 83\ mAh$, a little under 3% of a 3000 mAh pack.

Practice: Constant Current

1. An intake motor draws 1.5 A at 12 V and runs for a total of 60 s during a match.

- What power does it use?

- How much energy does it use, in joules and in watt-hours?

- How much charge flows from the battery in mAh?

2. During the 120 s driver-controlled period, each of your four drive motors averages 2.5 A, and the battery holds at about 12.5 V.

- What is the total current drawn by the drivetrain?

- How many mAh does the drivetrain use in one driver-controlled period?

3. A full 3000 mAh battery has a nominal voltage of 12 V.

- How much energy does it store in Wh?

- Using your answer to number 2, estimate how many driver-controlled periods the drivetrain alone could run on one battery.

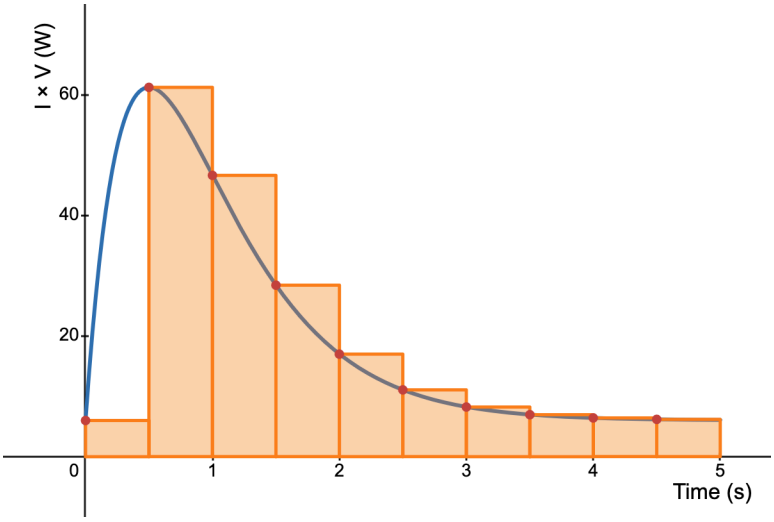
- Experienced teams swap in a fresh battery long before this number. Predict why, then read the Why Battery Voltage Drops section to check your prediction.

When Current and Voltage Change

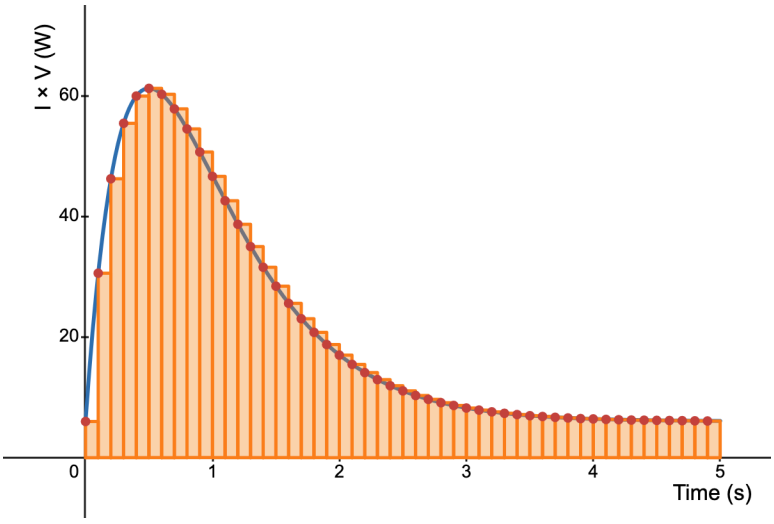
Real motors almost never draw constant current. A motor pulls a large spike when it starts or stalls, less as it speeds up, and a small holding current when it is barely loaded. At the same time, the battery voltage sags during each spike. When I and V both change, you can no longer multiply one current by one voltage by one time.

Instead, break the time into small intervals where I and V are close to constant, find the energy in each interval, and add them up:

$$E \approx \sum V_i \times I_i \times \Delta t$$



Each product $V_i \times I_i$ is the power during one interval, so each term $V_i \times I_i \times \Delta t_i$ is the area of a rectangle on a graph of power versus time. Adding the rectangles approximates the area under the power curve. As the time intervals get smaller, the sum gets closer to the exact area.



If you've taken integral calculus you will recall that this is the integral of the power curve with respect to time:

$$E = \int V_i \times I_i \, dt$$

When logging this data from an FTC robot you can typically get the time intervals down to 10-20 ms (0.01 - 0.02 s). This gives you a very fine approximation of the energy.

The same idea can be used to find the total charge drained from a battery.

$$Q \approx \sum I_i \times \Delta t$$

gives total charge, the area under the current versus time graph.

You can collect this data from your own robot. In the FTC SDK,

```java
// In initialization
DcMotorEx motor = hardwareMap.get(DcMotorEx.class, "flywheel"); VoltageSensor battery =
hardwareMap.voltageSensor.iterator().next();
ElapsedTime timer = new ElapsedTime();

// Every loop
double v = battery.getVoltage();
double i = motor.getCurrent(CurrentUnit.AMPS);
double t = timer.seconds();
```

Practice

4. An arm lifts a game element and then holds it in place. The motor's behavior has three phases:

Phase	**Duration (s)**	**Current (A)**	**Battery voltage (V)**
Start-up	0.5	8.0	11.9
Lifting	1.5	3.0	12.4
Holding	3.0	1.5	12.6

- Find the energy used in each phase and the total energy for the move.

- Find the total charge used, in coulombs and in mAh.

- Which phase uses the most energy? Is that the phase you expected? Explain.

5. A student logged this data once per second while a flywheel motor spun up and settled:

Time (s)	**Voltage (V)**	**Current (A)**	**Power (W)**
0	12.9	0.4	
1	12.1	9.0	
2	12.4	4.0	
3	12.5	2.5	
4	12.5	2.5	
5	12.6	1.0	

- Calculate the power at each time.

- Use Desmos or another graphing tool to graph the power vs. time.

- Estimate the total energy from 0 s to 5 s using rectangles, taking each interval's power from the start of the interval.

- Estimate it again using trapezoids, instead of rectangles.

- The two estimates differ. Which do you trust more, and why? What would you change about the data collection to get a better answer?

6. A teammate estimates the same energy from number 5 by averaging the six current readings, multiplying by 12 V, and multiplying by 5 s. Compare this to your answers in number 5. When would this shortcut work well, and when would it mislead you?

Why Battery Voltage Drops

It is tempting to think of voltage as "how much energy is left in the battery." That is not quite right. Voltage is energy per unit of charge: 1 V = 1 J/C. It tells you how much energy each unit of charge delivers as it moves through the circuit. It does not tell you how much charge the battery can still deliver.

Think of a water tower. The height of the water sets the pressure at the faucet and the amount of water in the tank sets how long the faucet can run. The “pressure” in a circuit is the voltage and the amount of water is the total charge in the battery. Total stored energy depends on both. A car battery and an FTC battery are both about 12 V, but a typical car battery holds roughly 50,000 mAh compared to the robot battery’s 3,000 mAh, so it stores more than 15 times as much energy at the same voltage.

So why does the voltage drop as the battery runs down? Two separate effects are at work.

Effect 1: Chemistry (state of charge). Each cell's voltage comes from the chemical reaction inside it, and the voltage depends on how much unreacted material is left at each electrode. A freshly charged NiMH cell rests at about 1.3 to 1.4 V. As the reaction proceeds, the voltage falls slowly through most of the discharge, then drops steeply near the end when little reactant remains. Ten cells in series make the 12 V nominal pack (10 × 1.2 V).

Effect 2: Internal resistance (voltage sag). Every battery has some internal resistance, $\mathbf{R_{internal}}$. When current flows, part of the voltage is used pushing current through the battery itself, so the voltage at the terminals is $\mathbf{V_{robot} = V_{rest} - I \times R_{internal}}$ where $\mathbf{V_{rest}}$ is the voltage measured with nothing drawing current from the battery. More current means more of the voltage is used for the internal current ($\mathbf{I \times R_{internal}}$) and less remains for the robot. Internal resistance also rises as a battery gets low, cold, or old. The power lost to internal resistance, $I^2 \times R_{internal}$, heats the battery instead of moving your robot.

This matters on the field. A motor's speed is roughly proportional to the voltage it receives. A robot with a sagging or low battery drives slower and lifts with less force, and an autonomous routine tuned on a fresh battery may come up short on a tired one. If the voltage sags far enough under a heavy load, the control electronics can brown out and reset.

Practice: Voltage and Energy

7. A battery reads 13.8 V with the robot idle. When the drivetrain pulls 10 A, the voltage drops to 12.3 V.

- Calculate the battery's internal resistance.

- Predict the robot voltage when the robot pushes against another robot and draws 20 A.

- How much power is wasted as heat inside the battery at 20 A?

8. The table below shows illustrative voltages for a 12 V, 3000 mAh NiMH pack at different states of charge, measured at rest and while the drivetrain draws about 15 A. Real values vary with the battery, its age, and the temperature.

State of charge	Charge remaining (mAh)	Voltage at rest (V)	Voltage while driving, about 15 A (V)
100%	3000	13.8	12.6
80%	2400	13.1	11.8
60%	1800	12.9	11.5
40%	1200	12.7	11.2
20%	600	12.4	10.6
5%	150	11.8	9.6
0%	0	11.0	7.5

- How much did the resting voltage fall between 80% and 40%? Between 20% and 5%?

- A teammate checks the battery, sees 12.7 V, and says it is "basically dead because it is below 13." Use the table to explain what is wrong with that reasoning.

- Why is a voltage reading alone a poor gauge of remaining energy in the middle of the discharge?

- Compare the two voltage columns. Why does the gap between them grow as the battery empties? Which of the two effects in the reading explains the gap?

- A robot's battery reads 7.5 V during a hard push late in a match, but 11 V a minute later in the pits. Use the table to explain how both readings can come from the same battery.

9. Write two or three sentences for a new team member explaining the difference between a battery's voltage and its stored energy. Use the words charge, voltage, and energy correctly.